

Solution to Fick's Equation

$$\frac{C(x,t) - C(\infty,t)}{C(0,t) - C(\infty,t)} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$C(\infty,t) = C_0 = 0.25 \text{ wt\% } C$$

$$C(0,t) = C_s = 1.2 \text{ wt\% } C$$

$$D = 1.6 \times 10^{-11} \frac{m^2}{s}$$

$$t = 10 \text{ s}$$

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x=0:5e-6:200e-6;
t=10;
D=1.6e-11;
y1=1-erf(x/(2*sqrt(D*t)));
Cx=0.25+y1*(1.2-0.25);
plot(x,Cx);
xlabel('x');
ylabel('C');
hold on;
t=100;
y2=1-erf(x/(2*sqrt(D*t)));
Cx1=0.25+y2*(1.2-0.25);
plot(x,Cx,x,Cx1);
t=1000;
y3=1-erf(x/(2*sqrt(D*t)));
Cx2=0.25+y3*(1.2-0.25);
plot(x,Cx,x,Cx1,x,Cx2);
legend('t=10s','t=100s','t=1000s');
title('Diffusion Ficks 2nd Law')
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